

RentalHub

Equipment Rental Platform — Project Architecture

1. Problem Statement

Traditional equipment rental processes often depend on phone calls, messages, manual records, and spreadsheets. This makes equipment discovery, price comparison, availability checking, booking, rental tracking, and return management difficult. Owners also need a simpler way to manage equipment, pricing, availability, and bookings. RentalHub solves these problems through a centralized web-based equipment rental platform.

2. Proposed Solution

RentalHub provides a single digital platform where customers can discover equipment, compare rental options, check availability, book equipment for specific dates, track rentals and returns, and submit reviews. Equipment owners can manage listings, pricing, availability, and bookings, while administrators can manage users, equipment, categories, and platform activities.

3. Core Technology Architecture

USERS Customer • Equipment Owner • Administrator
REACT Frontend / User Interface
NODE.JS + EXPRESS.JS Backend / REST API / Business Logic
MONGODB Database / Rental Data

Architecture flow: Customer / Owner / Admin → React → Express REST API → Node.js → MongoDB → Response to React

4. User Roles

Role	Key Functions
Customer	Browse equipment, search/filter, compare options, check availability, book for dates, track rental/returns
Equipment Owner	Create/manage equipment listings, set pricing, manage availability, manage bookings, and monitor equipment
Administrator	Manage users, equipment, categories, bookings, and platform activities; support quality control and

5. Core Functional Modules

Module	Purpose
Authentication	Registration, login, and role-based access.

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Equipment Management	Add, view, update, and remove equipment listings.
Category Management	Organize equipment into categories.
Search & Comparison	Help customers find and compare equipment and rental options.
Availability Management	Check date-based availability and prevent overlapping bookings.
Booking System	Create and manage bookings for selected dates.
Rental & Return Tracking	Track pickup, active rental, and return status.
Reviews	Allow customers to provide ratings and feedback.
Analytics	Use MongoDB aggregation for revenue and equipment utilization insights.

6. MongoDB Data Architecture

Collection	Purpose
Users	Customer, owner, administrator information and roles.
Categories	Equipment category information.
Equipment	Equipment details, owner, category, price, images, and availability.
Bookings	Customer, equipment, rental dates, amount, and booking status.
Rentals	Pickup, active rental, and return information.
Reviews	Customer ratings and comments for equipment.

7. Booking & Availability Flow

Step	Process
1	Customer selects equipment and rental start/end dates.
2	React sends the request to the Express API.
3	Node.js/Express checks MongoDB for overlapping bookings.
4	If dates are available, the booking is created.
5	Those dates become unavailable for that equipment.
6	Customer and owner can view booking/rental status.
7	After return, the status is updated and dates become available again.

8. Problem → Solution Mapping

Problem	RentalHub Solution
Manual rental records	Centralized digital rental lifecycle.
Double booking / scheduling conflicts	Date-based centralized availability checking.
Difficult equipment discovery	Searchable equipment marketplace.
Difficulty comparing rental options	Equipment details, pricing, and availability comparison.
Difficult rental tracking	Booking, rental, and return status tracking.
Owner management overhead	Equipment, pricing, availability, and booking management.
Limited business insights	MongoDB aggregation for revenue and utilization analysis.
Lack of centralized feedback	Customer reviews and ratings.

9. High-Level Project Flow

Customer: Browse → Check Availability → Compare → Select Dates → Book → Track Rental → Return → Review

Owner: Add Equipment → Set Price → Manage Availability → Receive Booking → Manage Rental → Track Utilization

Admin: Manage Users → Manage Equipment → Manage Categories → Monitor Bookings → Manage Platform Activities

10. Architecture Objective

RentalHub uses a three-layer architecture: React handles the user interface, Node.js with Express.js handles backend APIs and business logic, and MongoDB stores application data. This separation supports maintainability, centralized booking and availability management, and scalable rental operations.

11. Jury-Specified Core Stack

React + Node.js + Express.js + MongoDB are the primary technologies used to build RentalHub. The project features—authentication, CRUD, booking, availability, file handling, reviews, and MongoDB aggregation—are implemented around this core stack.